MODULE 03 · DOMAIN 1 - PLAN AND MANAGE AN AZURE AI SOLUTION

Choosing the service

Generation, grounding, search, agent tools and memory, and multimodal processing: which Foundry service does which job.

DOMAIN 1 - Plan and manage WEIGHT 25-30% MODULE 03 of 39

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The model generates. The services around it decide **what it knows**, **what it can do**, and **what it remembers**.

Five jobs, and the service for each

JOB	FOUNDRY SERVICE
Generate text, code or structured output	A Foundry model via the Responses API
Ground answers in your organisation's data	Foundry IQ, or file search for one agent's files
Find content by meaning, not exact words	Azure AI Search with an embedding model
Plan, call tools and act over several steps	Foundry Agent Service or Agent Framework
Understand images, audio, video or documents	A multimodal model, or Content Understanding

ON THE EXAM

Most questions in this area are one of these five jobs in disguise. Name the job first, then choose within it.

Retrieval or fine-tuning?

RETRIEVAL-AUGMENTED GENERATION

An open-book exam

- Adds knowledge the model never saw
- Private or frequently changing data
- Answers can cite their sources
- Update the documents, not the model

FINE-TUNING

Revising before the exam

- Changes behaviour, style or format
- Better performance on a specific task
- Not the way to add fresh facts
- Retrain when requirements change

THE TIEBREAKER

If the data changes weekly, or answers must cite documents, it is retrieval. Fine-tuning does not keep facts current.

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Three ways to give an agent knowledge

OPTION	WHAT IT IS	BEST WHEN
File search tool	Files uploaded to a vector store the agent searches	One agent, a set of files you provide
Azure AI Search tool	The agent queries an index you already run	An existing, well-tuned index
Foundry IQ knowledge base	Many sources behind one shared, permission-aware endpoint	Several agents, several sources, per-user access

THE SIGNAL

Many agents needing the same knowledge, with each user seeing only what they may see: Foundry IQ.

What a knowledge base actually does

- A **knowledge base** groups **knowledge sources**: Blob Storage, SharePoint, OneLake, the web, search indexes.
- **One knowledge base can serve many agents**, so retrieval is not rebuilt for each one.
- **Permission-aware:** syncs access lists, honours Purview labels, can query as the calling user.
- Returns extracted content **with citations**, so agents can reason over it and show sources.
- Built on **Azure AI Search**. You still need a search service, and its costs apply.

> ON THE EXAM
>
> Foundry IQ does not replace Azure AI Search. It is a managed layer on top of it.

Classic RAG or agentic retrieval?

CLASSIC RAG

One query, then the model

- A single hybrid query with semantic ranking
- Model used only to write the final answer
- Lower latency and fewer tokens
- Fits existing solutions with fewer code changes

AGENTIC RETRIEVAL

Plan, then search in parallel

- A model breaks the question into subqueries
- Uses chat history for follow-up questions
- Runs subqueries in parallel, then reranks
- Adds latency, but handles complex questions

THE TIEBREAKER

Microsoft recommends agentic retrieval for new RAG work. Simple, latency-critical lookups that already work can stay classic.

Keyword, vector, hybrid and semantic ranking

METHOD	GOOD AT	WEAK AT
Keyword	Exact terms: part numbers, error codes, names	Synonyms and paraphrase
Vector	Meaning: 'time off' finds 'annual leave'	Exact codes and rare identifiers
Hybrid	Both, with results merged into one list	Needs text and vector fields indexed
Semantic ranker	Re-ordering the top results by relevance	It reranks; it does not find new results

ON THE EXAM

Real users type error codes and paraphrases in the same week. That is why classic RAG uses hybrid plus semantic ranking.

Indexed or remote knowledge source?

INDEXED

Copied in, then searched

- Content is chunked, embedded and stored in an index
- Scheduled indexer runs keep it fresh
- Fast, rich keyword and vector queries
- Freshness depends on the refresh schedule

REMOTE

Queried where it lives

- Content is not copied into an index
- Searched live at the source
- The source enforces its own permissions
- Depends on the source's search quality

ON THE EXAM

Remote SharePoint sources are not indexed: SharePoint applies document permissions at query time.

Choosing agent tools

NEED	TOOL
Search the public web for current information	Web search
Run code: calculate, analyse a CSV, draw a chart	Code interpreter
Call logic that lives inside your own application	Function calling
Call an existing HTTP API described by a specification	OpenAPI tool
Use tools shared across agents, or owned by another team	MCP server
Hand work to another agent	Agent-to-agent (A2A), preview

ON THE EXAM

With function calling, the model only asks. Your application runs the function and returns the result.

Configure tools once, share them everywhere

- A **toolbox** bundles tools behind **one MCP-compatible endpoint**.
- Authentication, governance and versioning are managed in one place.
- Update a tool once, and every agent using the toolbox gets the change.
- Any MCP-capable runtime can use it: Agent Service, Agent Framework, LangGraph and others.
- A tool also needs support from both the **model** and the **region**.

> ON THE EXAM
>
> Twenty agents each configured with the same five tools is a toolbox scenario.

Conversation history or long-term memory?

CONVERSATION HISTORY

This conversation

- Earlier turns carried within one conversation
- Ends when the conversation ends
- Grows with every turn, costing tokens

MEMORY (PREVIEW)

Across sessions and devices

- Extracts user profile facts and chat summaries
- Stored in a managed memory store
- A scope keeps each user's memories separate
- Delete a scope to erase one user's data

WATCH FOR THIS

Memory stores facts about users. It is not a place to put company documents: that is a knowledge base.

Which service for non-text input?

REQUIREMENT	BETTER FIT
Describe a photo, or answer a question about it	A multimodal model
Extract fixed fields from forms, images, audio or video	Content Understanding
Screen uploaded images for harmful content	Azure Content Safety
Turn speech into text, or text into speech	Azure Speech

THE TIEBREAKER

Open-ended understanding: a model. The same fields, every time, in a schema: Content Understanding.

Old answers in this area that are expiring

FEATURE	STATUS AND REPLACEMENT
Azure OpenAI On Your Data	Retires 14 October 2026. Use Agent Service with Foundry IQ
Foundry workflows	Retire 1 December 2026. Use Agent Framework
Agents (classic)	Retire 31 March 2027. Use the new Agent Service
Prompt flow	Retires 20 April 2027. Use Agent Framework

ON THE EXAM

A large share of older RAG questions answer with On Your Data. On AI-103 the grounding answer is Foundry IQ.

> An insurer is building six agents, for claims, underwriting, HR and other teams. All need answers from a SharePoint policy library, PDFs in Blob Storage, and data in OneLake.
>
> Each employee must only see content they already have permission to see. The platform team wants to avoid building retrieval separately for every agent.

Which approach should the platform team use?

A Upload copies of the documents to each agent and attach the file search tool.

B Connect Azure OpenAI On Your Data to each source for every agent.

C Fine-tune one model on all the policy documents and share it across agents.

D One Foundry IQ knowledge base over all three sources, for every agent.

◆ PAUSE AND DECIDE

Which approach should the platform team use?

A Copies per agent with file search.

Duplicates content six times, goes stale, and loses each source's per-user permissions.

B Azure OpenAI On Your Data per source.

Deprecated and retiring in October 2026, and still configured separately for every agent.

C Fine-tune on the documents.

Fine-tuning does not reliably add facts, cannot enforce permissions, and goes stale.

D One Foundry IQ knowledge base for all agents.

Multiple sources, shared across agents, permission-aware, with citations.

THE SIGNAL

'Six agents', 'three sources' and 'only what they may see' together describe Foundry IQ almost word for word.

> A manufacturer's support assistant uses vector search over product manuals. Questions like 'why does the pump keep stopping' return excellent results.
>
> But when engineers type an error code such as E-4417, results often describe unrelated errors with similar wording.

What should the team change?

A Switch to hybrid search so keyword matching catches exact codes.

B Replace the embedding model with a larger, more accurate embedding model.

C Return twice as many chunks to the model with every query.

D Store the error code tables in the assistant's long-term memory.

◆ PAUSE AND DECIDE

What should the team change?

A Switch to hybrid search.

Keyword matching finds exact codes; vectors keep handling paraphrase. Both are merged.

B Larger embedding model.

Still matches on meaning. Rare identifiers stay weak, at extra cost.

C Return more chunks.

More of the wrong results, more tokens, and more noise for the model.

D Put code tables in memory.

Memory holds facts about users, not reference documents.

THE SIGNAL

Paraphrase works, exact codes fail: that split is the fingerprint of vector-only search.

> A travel company's booking agent should remember each traveller's seat preference and dietary needs, on future trips and on any device.
>
> Under data protection law, the company must be able to erase an individual traveller's remembered details on request.

What should the company use?

A. Index every past conversation into a Foundry IQ knowledge base for the agent.

B. Send each traveller's full conversation history at the start of every new session.

C. Agent memory with one scope per traveller, deleting that scope on request.

D. Fine-tune the agent's model each month on the latest booking conversations.

◆ PAUSE AND DECIDE

What should the company use?

A Index conversations into Foundry IQ.

A knowledge base is for shared documents. Every traveller's chats in one index is a privacy risk.

B Replay full history each session.

Token cost grows with every trip, and it spreads raw personal data around.

C Memory, one scope per traveller, delete on request.

Extracts preferences across sessions and devices; deleting a scope erases one person's data.

D Monthly fine-tuning on conversations.

Bakes personal data into the model, where it cannot be erased for one person.

PREVIEW NOTE

Memory is preview, but it is the platform's answer to 'remember this person, and forget them on request'.

A retailer's shopping app uses classic RAG, with hybrid search and semantic ranking. Questions are short single-product lookups, and answers are accurate.

Retrieval has a strict latency budget. A new architect proposes moving to agentic retrieval because Microsoft recommends it for new RAG work.

What should the team decide?

A Migrate, because agentic retrieval improves the accuracy of every kind of workload.

B Keep classic RAG: simple queries, critical latency, and it already works.

C Replace retrieval with a model fine-tuned on the product catalogue.

D Switch from hybrid to vector-only search to cut latency further.

◆ PAUSE AND DECIDE

What should the team decide?

A. Migrate for accuracy everywhere.
Agentic retrieval helps complex, multi-part questions, and adds latency these lookups cannot afford.

B. Keep classic RAG.
Simple queries, a strict latency budget and accurate results: the case Microsoft describes for classic.

C. Fine-tune on the catalogue.
Prices and stock change constantly. Fine-tuning cannot keep product facts current.

D. Vector-only search.
Loses exact matching on product names and codes, trading accuracy for little gain.

THE TIEBREAKER

'Recommended for new RAG work' is not 'mandatory for all RAG work'. Read who the recommendation is for.

Choosing services in five lines

- **Knows, does, remembers:** grounding, tools and memory fix different gaps.
- **Retrieval adds knowledge; fine-tuning changes behaviour.**
- **Many agents, many sources, per-user access:** Foundry IQ. One agent's files: file search.
- **Hybrid** covers codes and paraphrase. **Agentic retrieval** for complex questions, at some latency.
- **Memory is for facts about users**, scoped per user. On Your Data is retiring.

> ON THE EXAM
>
> Next, module 04: the infrastructure and deployment options these services run on.